

31/1/24

Date.....

Biology

1. (4) b and d

2. (4) All of these

3. (1) Tallness is dominant over dwarfness

4. (3) b, c

5. (3) Translocation

6. (1) Arteries

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Ans 21- Raw materials for photosynthesis are carbon dioxide and water.

They get

Carbon dioxide (CO_2) $\longrightarrow$ Stomata on leaves.

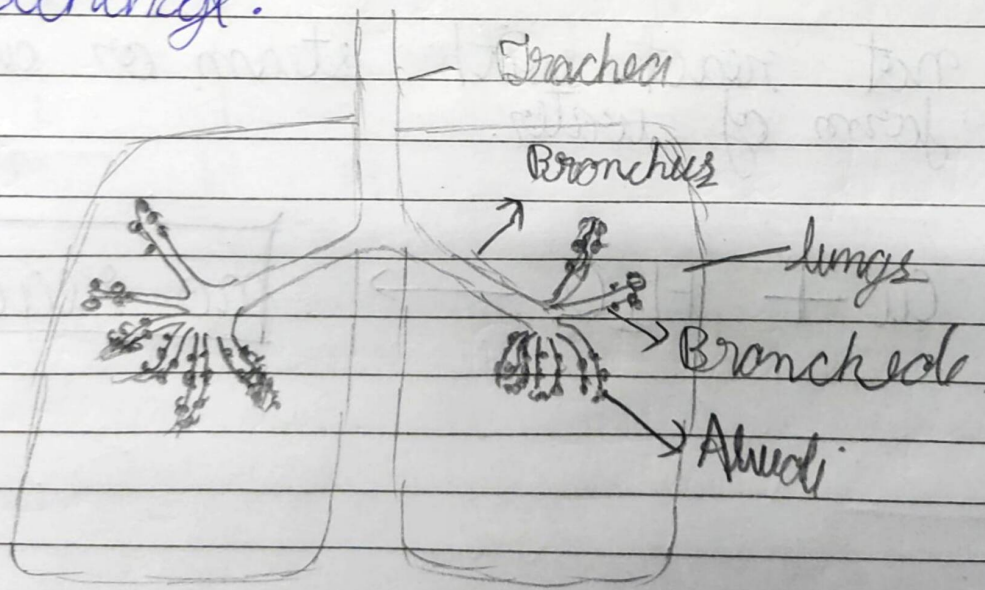
Water (H_2O) $\longrightarrow$ Roots by the process of osmosis

22- Haemoglobin in our blood is responsible for respiration as it carries oxygen from lungs to the

Ans 27- Lungs are designed in human beings for the intake of oxygen to for the process of respiration.
ma

When we take in oxygen. The oxygen goes to trachea then bronchus then bronchiole.

At the end of bronchioles there is a sack like structure called alveoli which helps in exchange of gases by ~~and~~ increasing the surface area for gaseous ~~exchange~~ exchange.



Ans 28 -

(a) The functions of liver and pancreas are as follows

Liver → • helps in breakdown of larger food molecules.
• Releases bile juice.

Pancreas → • Releases digestive juices like lipase which helps in emulsification of fat.

Ans 28.(b) Nutrition

The process of taking in of food and utilising it for growth, repair and development of our body is known as nutrition.

Nutrients

The food which we eat breaks down for its utilisation in the form of nutrients.

The differences ~~part~~ between holozoic nutrition and saprophytic nutrition are as follows:-

Holozoic	Saprophytic
① Get their food from plants or animals.	① Get their food from <u>dead and decaying organism</u> .
② Can be <u>omnivorous</u>	② They are <u>carnivorous</u> .
③ For eg. Humans, Cow	③ For eg. tiger <u>tiger decomposers</u> , bacteria.

Ans 29 - The differences are as follows

Self Pollination

① Only one plant is needed.

② The one plant involved contains both female and male reproductive system.

③ ~~Bisexual~~ Unisexual

Gross Pollination

① Two plants are necessary.

② In the two plants one will contain male reproductive part and one will contain female reproductive system.

③ Bisexual.

These are the various agents which helps in pollination:

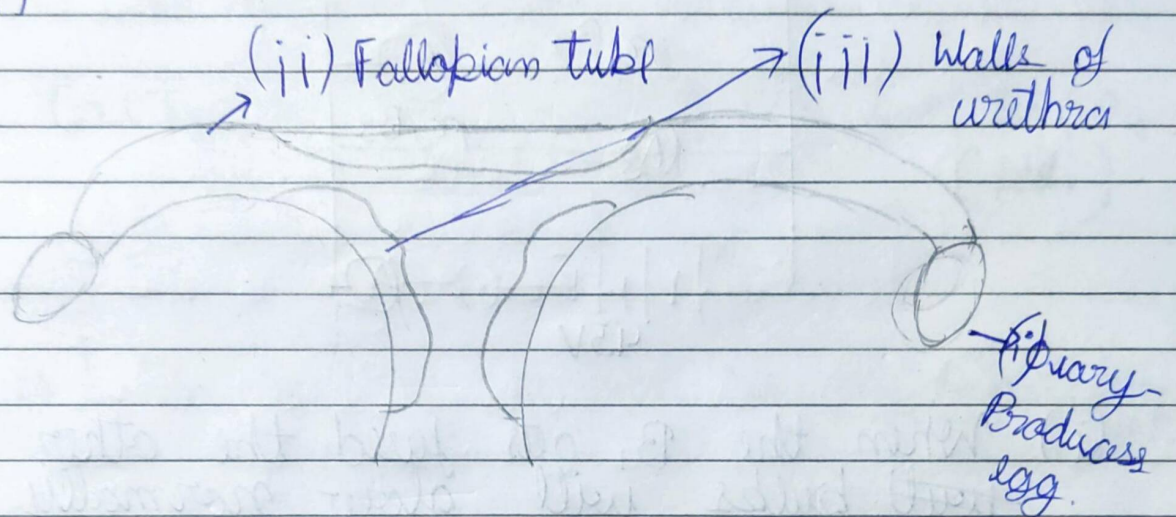
① Water

② Butterfly

③ Wind.

Ans 34-

(i)



~~(ii) When the human egg is not fertilised~~

(ii) In females ovary produces eggs every month. When the egg is not fertilised the egg breaks down with breaking down of walls of urethra and comes out from vagina. This process is known as menstruation or periods.

Ans 37-

(i) Mendel selected a pea plant for his experiments as

- ① It has a shorter life span and can be grown ~~fast~~ faster helping in observing various generations
- ② It gives seeds directly and can be observed with naked eye easily.

(ii) Dominant is the trait which appears physically in the ~~progeny~~ F_1 progeny. Recessive is the trait which is suppressed by the dominant trait in the F_1 generation.

For eg. If we take a Tall pea plant (TT) which is homozygous and short & homozygous plant (tt).

Then the F_1 generation will have all the Tall plants as the trait of tallness is dominant.

F_1 generation $\rightarrow$

$P \rightarrow$	T	t
T	TT	Tt
t	Tt	tt

$\therefore$ All the plant will be tall.

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(iii) These are the two visible traits of garden pea that Mendel considered in his experiment: -

① Colour of the seed.
eg. green and yellow

② Shape of the seed.
eg. Round and wrinkled